

pipeline

https://huggingface.co/docs/transformers/main/en/main_classes/pipelines#transformers.pipeline

Detailed Documentation

`<span class="code-`

`inline">transformers.pipeline``transformers.pipeline``https``:/`

```
transformers.pipelinehttps://github.com/huggingface
"task", "val": ": typing.Optional[str] = None"}, {"name": "model", "val": ":
typing.Union[str, ForwardRef('PreTrainedModel'), NoneType] = None"}, {"name":
"config", "val": "": typing.Union[str,
transformers.configuration_utils.PreTrainedConfig, NoneType] = None"},
{"name": "tokenizer", "val": "": typing.Union[str,
transformers.tokenization_utils.PreTrainedTokenizer,
ForwardRef('PreTrainedTokenizerFast'), NoneType] = None"}, {"name":
"feature_extractor", "val": "": typing.Union[str,
ForwardRef('SequenceFeatureExtractor'), NoneType] = None"}, {"name":
"image_processor", "val": "": typing.Union[str,
transformers.image_processing_utils.BaseImageProcessor, NoneType] =
None"}, {"name": "processor", "val": "": typing.Union[str,
transformers.processing_utils.ProcessorMixin, NoneType] = None"},
{"name": "revision", "val": ": typing.Optional[str] = None"}, {"name": "use_fast",
"val": ": bool = True"}, {"name": "token", "val": ": typing.Union[str, bool,
```

```

NoneType] = None"}, {"name": "device", "val": ": typing.Union[int, str,
ForwardRef('torch.device'), NoneType] = None"}, {"name": "device_map", "val":
": typing.Union[str, dict[str, typing.Union[int, str]], NoneType] = None"}, {"name":
"dtype", "val": ": typing.Union[str, ForwardRef('torch.dtype'), NoneType] =
'auto'"}, {"name": "trust_remote_code", "val": ": typing.Optional[bool] = None"},
{"name": "model_kwargs", "val": ": typing.Optional[dict[str, typing.Any]] = None"},
{"name": "pipeline_class", "val": ": typing.Optional[typing.Any] = None"},
{"name": "**kwargs", "val": ": typing.Any"}]- **task** ( str ) --

```

The task defining which pipeline will be returned. Currently accepted tasks are:

- `"audio-classification"`: will return a [AudioClassificationPipeline]
(/docs/transformers/main/en/main_classes/pipelines# `transformers.AudioCL`
- `"automatic-speech-recognition"`: will return a
[AutomaticSpeechRecognitionPipeline]
(/docs/transformers/main/en/main_classes/pipelines# `transformers.Automat`
- `"depth-estimation"`: will return a [DepthEstimationPipeline]
(/docs/transformers/main/en/main_classes/pipelines# `transformers.DepthEs`
- `"document-question-answering"`: will return a
[DocumentQuestionAnsweringPipeline]
(/docs/transformers/main/en/main_classes/pipelines# `transformers.Documen`
- `"feature-extraction"`: will return a [FeatureExtractionPipeline]
(/docs/transformers/main/en/main_classes/pipelines# `transformers.Feature`
- `"fill-mask"`: will return a [FillMaskPipeline]
(/docs/transformers/main/en/main_classes/pipelines# `transformers.FillMas`
- `"image-classification"`: will return a [ImageClassificationPipeline]

(/docs/transformers/main/en/main_classes/pipelines# `transformers.ImageCL`

- `"image-feature-extractio`

`class <span class="code-`

`inline">transformers.AudioClassificationPipelinetransformers.Auc`

`class`

```
transformers.AudioClassificationPipelinetransformers.AudioClassificati
**args", "val": ""}, {"name": "**kwargs", "val": ""}]- **model** ([PreTrainedModel]
(/docs/transformers/main/en/main_classes/model# transformers.PreTrainedMoc
--
```

The model that will be used by the pipeline to make predictions. This needs to be a model inheriting from

`[PreTrainedModel]`

(/docs/transformers/main/en/main_classes/model# `transformers.PreTrainedMoc`

- `**feature_extractor**` (`[SequenceFeatureExtractor]`
(/docs/transformers/main/en/main_classes/feature_extractor# `transformers.`
--

The feature extractor that will be used by the pipeline to encode data for the model. This object inherits from

`[SequenceFeatureExtractor]`

(/docs/transformers/main/en/main_classes/feature_extractor# `transformers.Seqi`

- `**modelcard**` (str or ModelCard, *optional*) --

Model card attributed to the model for this pipeline.

- `**task**` (str, defaults to "") --

A task-identifier for the pipeline.

`__call__` transformers.AudioClassificationPipeline.__call__ https://github.com/huggingface/transformers/blob/master/src/transformers/pipelines/audio_classification.py

`__call__` transformers.AudioClassificationPipeline.__call__ https://github.com/huggingface/transformers/blob/master/src/transformers/pipelines/audio_classification.py
 "inputs", "val": ": numpy.ndarray | bytes | str | dict"}, {"name": "**kwargs", "val": ": typing.Any"}]- `**inputs**` (np.ndarray or bytes or str or dict) --

The inputs is either :

- str that is the filename of the audio file, the file will be read at the correct sampling rate

to get the waveform using `*ffmpeg*`. This requires `*ffmpeg*` to be installed on the system.

- bytes it is supposed to be the content of an audio file and is interpreted by `*ffmpeg*` in the
- (np.ndarray of shape (n,) of type np.float32 or np.float64)

Raw audio at the correct sampling rate (no further check will be done)

- dict form can be used to pass raw audio sampled at arbitrary

sampling_rate and let this

pipeline do the resampling. The dict must be either be in the format
{"sampling_rate": int,

"raw": np.array} , or {"sampling_rate": int, "array": np.array} , where the key
"raw" or

class <span class="code-

inline">transformers.AutomaticSpeechRecognitionPipelinetransf

class

```
transformers.AutomaticSpeechRecognitionPipelinetransformers.AutomaticS
"model", "val": ": PreTrainedModel"}, {"name": "feature_extractor", "val": ":
typing.Union[ForwardRef('SequenceFeatureExtractor'), str, NoneType] = None"},
{"name": "tokenizer", "val": ":
transformers.tokenization_utils.PreTrainedTokenizer | None = None"},
{"name": "decoder", "val": ":
typing.Union[ForwardRef('BeamSearchDecoderCTC'), str, NoneType] = None"},
{"name": "device", "val": ": typing.Union[int, ForwardRef('torch.device'),
NoneType] = None"}, {"name": "**kwargs", "val": ""}]- **model**
([PreTrainedModel]
(/docs/transformers/main/en/main_classes/model# transformers.PreTrainedMoc
--
```

The model that will be used by the pipeline to make predictions. This needs to be
a model inheriting from

[PreTrainedModel]

(/docs/transformers/main/en/main_classes/model# transformers.PreTrainedModel)

- **feature_extractor** ([SequenceFeatureExtractor]
(/docs/transformers/main/en/main_classes/feature_extractor# transformers.
--

The feature extractor that will be used by the pipeline to encode waveform for the model.

- **tokenizer** ([PreTrainedTokenizer]
(/docs/transformers/main/en/main_classes/tokenizer# transformers.PreTrained
--

The tokenizer that will be used by the pipeline to encode data for the model. This object inherits from

[PreTrainedTokenizer]

(/docs/transformers/main/en/main_classes/tokenizer# transformers.PreTrained

- **decoder** (pyctcdecode.BeamSearchDecoderCTC, *optional*) --

BeamSearchDecoderCTC](https://github.com/kensho-technologies/pyctcdecode/blob/2fd33dc37c4111417e08d89ccd23d28e9b308d19

__call__ transformers AutomaticSpeechRecognitionPipeline.__call__

__call__ transformers AutomaticSpeechRecognitionPipeline.__call__ https://github.
"inputs", "val": ": numpy.ndarray | bytes | str | dict"}, {"name": "***kwargs", "val": ":

typing.Any"]]- **inputs** (np.ndarray or bytes or str or dict) --

The inputs is either :

- str that is either the filename of a local audio file, or a public URL address to download the

audio file. The file will be read at the correct sampling rate to get the waveform using

ffmpeg. This requires *ffmpeg* to be installed on the system.

- bytes it is supposed to be the content of an audio file and is interpreted by *ffmpeg* in the
- (np.ndarray of shape (n,) of type np.float32 or np.float64)

Raw audio at the correct sampling rate (no further check will be done)

- dict form can be used to pass raw audio sampled at arbitrary sampling_rate and let this

pipeline do the resampling. The dict must be in the format {"sampling_rate": int, "raw":

class <span class="code-

inline">transformers.TextToAudioPipelinetransformers.TextToAu

class

transformers.TextToAudioPipelinetransformers.TextToAudioPipelinehttps

```
"*args", "val": ""}, {"name": "vocoder", "val": " = None"}, {"name": "sampling_rate", "val": " = None"}, {"name": "no_processor", "val": " = True"}, {"name": "**kwargs", "val": ""}]
```

Text-to-audio generation pipeline using any `AutoModelForTextToWaveform` or `AutoModelForTextToSpectrogram`. This

pipeline generates an audio file from an input text and optional other conditional inputs.

Unless the model you're using explicitly sets these generation parameters in its configuration `files` (`generation_config.json`), the following default values will be used:

- `max_new_tokens`: 256

`__call__` transformers.TextToAudioPipeline.__call__ <https://github.com>

`__call__` transformers.TextToAudioPipeline.__call__ <https://github.com/huggingface>
"text_inputs", "val": ": str | list[str]"}, {"name": "**forward_params", "val": ""}]-
text_inputs (`str` or `list[str]`) --

The `text` (s) to generate.

- `**forward_params**` (`dict` , *optional*) --

Parameters passed to the model generation/forward method. `forward_params` are always passed to the underlying model.

- `**generate_kwargs**` (`dict` , *optional*) --

The dictionary of ad-hoc parametrization of `generate_config` to be used for the generation call. For a

complete overview of generate, check the [following

guide]

(https://huggingface.co/docs/transformers/en/main_classes/text_generation).

`generate_kwargs` are

only passed to the underlying model if the latter is a generative model.0A `dict` or a list of `dict` The dictionaries have two keys:

class `<span class="code-`

inline">transformers.ZeroShotAudioClassificationPipelinetransfor

class

```
transformers.ZeroShotAudioClassificationPipelinetransformers.ZeroShotA
**kwargs",      "val":      "}}}-      **model**      ([PreTrainedModel]
(/docs/transformers/main/en/main_classes/model# transformers.PreTrainedMoc
--
```

The model that will be used by the pipeline to make predictions. This needs to be a model inheriting from

[PreTrainedModel]

(/docs/transformers/main/en/main_classes/model# transformers.PreTrainedModel)

- **tokenizer** ([PreTrainedTokenizer])

(/docs/transformers/main/en/main_classes/tokenizer# transformers.PreTrainedTokenizer)

--

The tokenizer that will be used by the pipeline to encode data for the model. This object inherits from

[PreTrainedTokenizer]

(/docs/transformers/main/en/main_classes/tokenizer# transformers.PreTrainedTokenizer)

- **feature_extractor** ([SequenceFeatureExtractor])

(/docs/transformers/main/en/main_classes/feature_extractor# transformers.SequenceFeatureExtractor)

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